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Optimization of Microbial Control Program & Integrity Augmentation– Transformation of Core Processes in Upstream Industry

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Abstract

Ensuring the integrity of crude oil network is of paramount importance for maintaining operational efficiency and minimizing environmental impact. Control of microbiologically induced corrosion (MIC) is critically essential for energy producers since MIC can progress at aggressive localized rates, leading to unpredictable and premature failures that result in loss of primary containment and disruptions in production. This paper details the MIC optimization program at ADNOC's largest oil- producing field, which ensured the network's integrity and resulted in a substantial OPEX saving of 81%, amounting to over 3 million USD/Year.

Microbial induced corrosion control through biocide application is always considered as a safe way to augment the integrity of network. Like many other operators, we deployed a rotational biocide scheme in batch treatments to avoid creating biocide resistant microbial population. This case study presents a comprehensive, analytical approach using extensive field data to evaluate the optimization of required biocide dosage in our crude oil network.

Implementation of MIC optimization program was implemented after careful review of below:

- MIC Damage Mechanism and Control
- MIC Risk Assessment Results for the crude oil network.
- Biocide Optimization Strategy.
- Biocide Optimization Roadmap and Recommendations

Through assessment of the crude oil network was performed and bench marked with world leading energy producers. From wellhead to export network, all the crude oil network was divided into categories based on the following parameters:

- Temperature, Nutrients and H₂S.
- Flow / velocity regime.

- Bacterial monitoring (cells / ml)
- Cumulative score, benchmarked with world leading Oil Producers.

Based on the calculated bench marking score cards, the whole network was divided in two categories:

Risk	Network
No Risk	• Wellhead pipe network and transfer lines to Stations. • Temperature 60 - 100 C • High salinity >250,000 mg/L • H₂S >0.2 bar • Sulfates 20- 300 mg/L • Velocity >3 m/sec • Bacterial monitoring no more than 100 cells/ml.
High Risk	• Crude Oil Export Network • Temperature < 60 C • Low salinity ~60,000 mg/L • No H₂S. • Sulfates 20 - 300 mg/L • Velocity 1-2 m/sec. • Bacterial monitoring > more than 1000 cells/ml.

Based on evaluation, MIC treatment was optimized and stopped for wellhead and transfer lines to station network with immediate effect, resulting in saving of more than 2407 drums / year equal to 2 MMUSD/ Year.

Biocide treatment for export network was optimized from 500 ppm to 300 ppm and from batch treatment of 5 hrs to 3 years, which resulted in optimization of 1490 drums equivalent to 1.05 MMUSD/ Year.

This novel implementation shows that the integrity of the crude oil network can be effectively maintained with an optimized MIC treatment plan. The bench marking and evaluation criteria presented in this paper can be utilized by other oil and gas producers to effectively manage MIC treatment plan and optimize OPEX significantly.

Moving forward, the integration of these optimized treatment plans will serve as a benchmark for future operations, inspiring other producers to adopt similar strategies. This commitment to sustainability and operational excellence ensures that we continue to protect our valuable resources and environment for future generations.

INTRODUCTION

Corrosion remains a persistent threat to oil and gas infrastructure, causing significant operational downtime, economic losses, and environmental hazards. Among the diverse mechanisms of corrosion, microbiological influenced corrosion (MIC) poses a unique challenge due to its unpredictable nature and the involvement of complex biological mechanism. This comprehensively explores different modes of corrosion in petroleum operations, effective treatment methodologies, and the critical role of microbial control programs. It places particular emphasis on the ADNOC Onshore initiative for the optimization of MIC dosage, presenting it as a benchmark for the sustainable operations and integrity management.

Oil and gas industry relies heavily on maintaining the structural and operational integrity of the pipelines network, reservoirs, and processing facilities. Corrosion is a natural but damaging process, threatens both the economic viability and sustainability of energy production. ADNOC Onshore, a leader in operational

excellence, has pioneered an integrated approach to corrosion control, especially related to MIC, achieving substantial cost savings and emission reductions. This paper investigates the scientific basis and practical implementation of such programs, guiding future strategies across the sector.

UNDERSTANDING CORROSION: MODES AND MECHANISMS

Uniform Corrosion

Uniform corrosion, also called as general attack, occurs evenly across the exposed surface. It results in a consistent thinning of the metal. It is often predictable and relatively straightforward to monitor. In oilfield environments, uniform corrosion frequently results from the prolonged exposure to oxygenated water or acidic conditions and leads to gradual degradation.

Localized Corrosion

Localized corrosion is highly aggressive in specific areas and can lead to the catastrophic failures. Major localized corrosion modes include:

- Pitting Corrosion: Small, deep holes form on the metal surfaces, often initiated by chloride ions in brine or impurities in the water phase.
- Crevice Corrosion: Occurs in the shielded areas such as gaskets, flanges, where localized chemistry supports the aggressive attack.

Galvanic Corrosion

Galvanic corrosion arises when two dissimilar metals are connected electrically in the presence of an electrolyte. The anode metal corrodes preferentially. In the oilfield environment, it is very common when carbon steel components contact stainless steel or copper-based alloys.

Erosion Corrosion

This mode of corrosion combines mechanical wear with the chemical attack, accelerated by high-velocity fluids laden with sand or other particulates. Erosion corrosion is common in pipelines with turbulent flow or at points where flow direction changes abruptly.

Stress Corrosion Cracking (SCC)

SCC occurs due to combined influence of tensile stress and corrosive environment, leading to cracking that can propagate rapidly and unpredictably. Common agents include sulfides, chlorides, and caustic substances, which are frequently encountered in oilfield operations.

Microbiologically Influenced Corrosion (MIC)

MIC is a complex process where microorganisms, especially bacteria, directly or indirectly accelerate the metal deterioration. Sulfate-reducing bacteria (SRB), iron-oxidizing bacteria, and acid-producing bacteria are among the primary culprits. MIC often leads to pitting, under-deposit corrosion, and unpredictable failures of pipeline network leading to loss of primary containment incidents.

IMPLICATIONS DUE TO MIC

Microbially induced corrosion (MIC) in the oil transfer lines leads to significant financial implications caused by increased maintenance, repairs, and potentially catastrophic failures. MIC, caused by microorganisms, accelerates metallic corrosion, which impacts pipeline integrity and reliability. Some studies suggest that MIC accounts for a substantial portion of global corrosion-related economic losses, potentially reaching \$250-\$500 billion annually. Moreover as per American Institute of Physics, 15-30% of pipeline corrosion in Oil and Gas sector is linked to MIC.

Increased Maintenance and Repair Costs

MIC causes more frequent inspections, pigging operation for cleaning, and repairs of the corroded segments, leading to higher operational costs.

Pipeline Replacement

MIC can result in significant damage, requiring a premature pipeline / piping replacement, which is a substantial capital expense and production deferment.

Production Downtime

Corrosion-related failures can cause unexpected long shutdowns, leading to loss of production and revenue.

Environmental Damage and Liability

Leaks from the corroded pipelines results in costly environmental impact and potential legal liabilities.

FUNDAMENTALS OF MIC

Microbiologically Influenced Corrosion (MIC) refers to the deterioration of metal surfaces resulting from environmental factors, compounded by the metabolic activities of microorganisms, particularly bacteria. Typically, components necessary for corrosion are present prior to the bacterial involvement. However, conditions conducive to the microbial corrosion can also foster corrosion independent of bacterial activity.

The presence and biological activity of microorganisms significantly accelerate and intensify corrosion processes. MIC is characterized as a biofilm-associated phenomenon, wherein bacteria implicated in corrosion are retained within the biofilm matrix. These bacteria depend on the collective functions of various microorganisms in the biofilm, and the accumulation of metabolic by-products substantially affects the type and extent of metal degradation observed. Identifying a single, unifying mechanism for MIC is improbable due to its complexity.

MIC entails the proliferation of microorganisms within engineered environments that provide essential conditions for specific bacterial species. This growth can occur through several mechanisms, such as oxygen consumption, hydrogen utilization, production of corrosive *metabolites*, formation of concentration cells, interaction with protective coatings and inhibitors, and depletion of metal ions. Sulfate-Reducing Bacteria (SRB) are among the most recognized agents responsible for MIC; however, their absence does not preclude the possibility of microbial corrosion, as other organisms—including iron-fixing bacteria, Acid-Producing Bacteria (APB), sulfide-generating bacteria, sulfur bacteria, Salmonella species, methanogens, and slime-forming bacteria—can induce severe corrosion beneath deposits.

Generally, microbial corrosion occurs in environments where pH values range from 5 to 9 and total suspended solids are below 200,000 ppm. Poorly dissolved solids do not inhibit bacterial proliferation, especially under the dynamic flow conditions. SRBs require as little as 10 ppm of sulfate to remain active, and the presence of oxygen does not impede the growth of anaerobic bacteria.

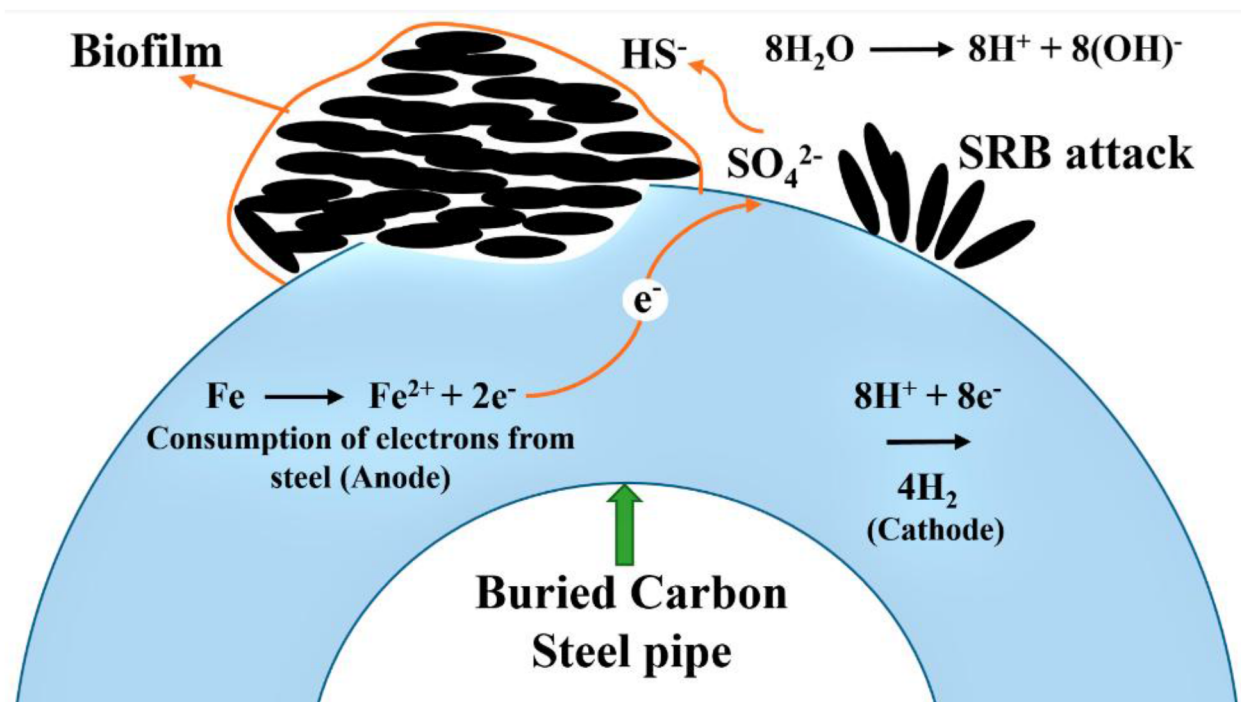


Figure 1—Chemical formulation



Figure 2—Metal affected by MIC

OILFIELD ENVIRONMENTS AND SOURING THEORY

An oil well creates an environment that undergoes gradual but continual change from the moment drilling starts. Although bacteria are likely introduced during the drilling phase, the initially low water content in the produced fluids generally leads to minimal microbial activity in the production wells.

With depletion of reservoir water cut increased especially after water breakthrough, the level of microbial activity correspondingly increases. The introduction of secondary recovery water further promotes the microbial growth within the producing wells.

Oil flowlines, manifolds, gathering stations, processing facilities, and disposal systems can experience microbial induced corrosion due to bacterial growth, particularly in low-lying areas or regions with stagnant flow. Water-flood sources often contain microorganisms and nutrients that promote further microbial activity. The mixing of injected waters with connate reservoir water supports additional microbial development. Various explanations for reservoir souring have been presented in the literature, though research suggests untreated waterflood injection is a main factor. Bacteria may proliferate and move with water from injector to producer locations. Hydrogen sulfide (H_2S), a metabolic byproduct, migrates more rapidly than bacterial biofilm. Therefore, detecting increased H_2S concentration at the producer is considered an indicator of reservoir souring, as depicted in the associated flow diagram.

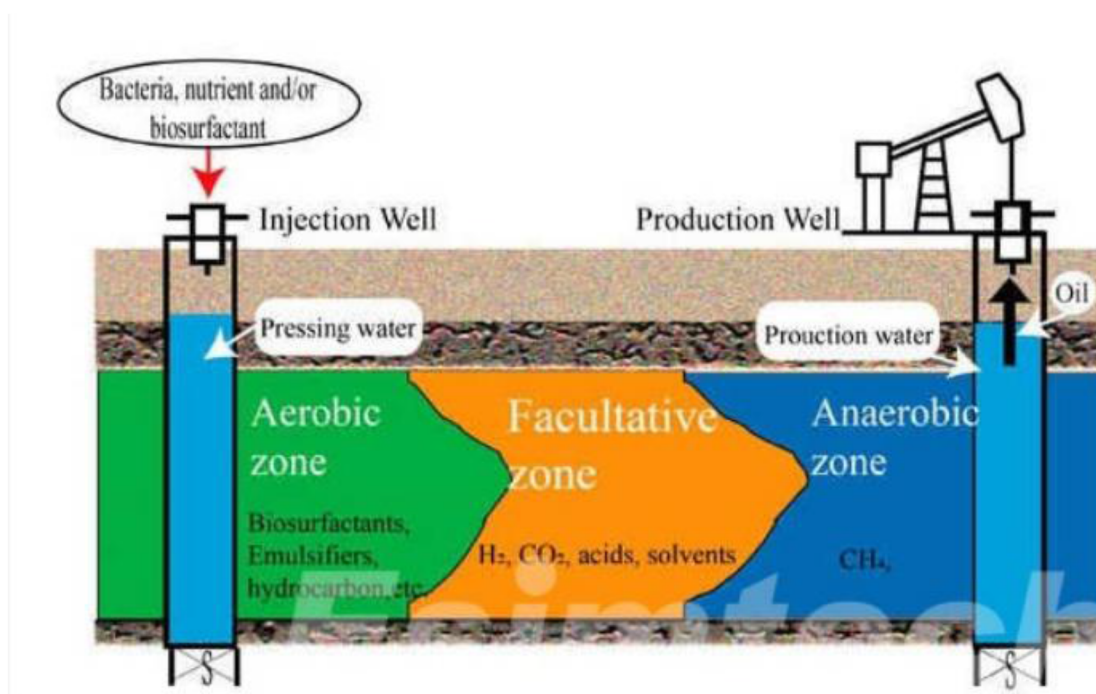


Figure 3—Bacterial mobilization from Water Injection well to Oil Producer

TREATMENT STRATEGIES FOR CORROSION CONTROL

Material Selection

Employing corrosion-resistant alloys, properly coated steels, or non-metallic materials can prevent or delay the onset of corrosion. Selection is based on the expected corrosive agents, operating temperatures, and mechanical stresses.

Protective Coatings and Linings

Inorganic and organic coatings such as epoxy, polyurethane, and metallic linings (e.g., cladding with stainless steel) act as barriers to corrosive agents. Coating integrity is maintained through regular inspection and timely repair.

Cathodic Protection

This electrochemical method uses sacrificial anodes or impressed current systems to protect the primary structure by making it the cathode of an electrochemical cell. Widely used for pipelines and storage tanks, cathodic protection is highly effective when properly designed and maintained.

Chemical Inhibitors

Corrosion inhibitors are chemicals injected into process streams to form protective films on metal surfaces or neutralize corrosive agents. Common inhibitors include amines, phosphates, and filming amines. The choice of inhibitor depends on the nature of the corrosive environment and compatibility with operational parameters.

Operational Practices

Regular monitoring, pigging (for pipelines), and process optimization can minimize the stagnant zones, turbulence, or deposit formation, all of which can promote the localized corrosion. Scheduled maintenance and inspection play a pivotal role.



Figure 4—Pigging of the Transfer line

Biocides and Microbial Control

Biocides are crucial for controlling the MIC related corrosion, as they target and neutralize problematic microbes. These are applied as continuous, batch, or rotational treatments, with the choice of biocide and application regime tailored to microbial diversity and resistance patterns.

THE IMPORTANCE OF MICROBIAL CONTROL PROGRAMS

MIC: A Hidden Threat

While general corrosion can often be anticipated and managed, MIC introduces unpredictability due to its dependence on dynamics of microbial populations. The oilfield environment, rich in water, nutrients, and hydrocarbons, provides an ideal habitat for the bacteria that accelerate corrosion.

Key Microbial Agents in Oilfields

- Sulfate-Reducing Bacteria (SRB): Produce hydrogen sulfide (H_2S), which is highly corrosive and can cause rapid pitting of steel.
- Iron-Oxidizing Bacteria: Accelerate corrosion by depositing iron oxides and creating localized differential aeration cells.
- Acid-Producing Bacteria (APB): Generate organic acids that lower pH and destabilize protective films.



Figure 5—Recovered sludge Samples

Program Components

A robust microbial control program must include:

- Regular microbial monitoring through sampling and laboratory analysis.
- Risk assessment to identify vulnerable network segments.
- Strategic biocide application (dose, frequency, rotation to prevent resistance).
- Data-driven optimization based on field results and benchmarking studies.

Advantages of Microbial Control Programs

- Prevention of unpredictable and catastrophic failures.
- Operational savings by targeting the treatments where it is needed the most.
- Long-term asset integrity, reducing unplanned shutdowns and repairs.
- Improved environmental compliance by minimizing leaks and emissions.

ADNOC'S APPROACH TO MIC OPTIMIZATION: A CASE STUDY IN EXCELLENCE

ADNOC Onshore's largest asset, with decades of aquifer water injection for reservoir pressure support, faced increasing volumes of produced water and environmental sustainability challenges. With corrosion—especially MIC—emerging as a critical risk, ADNOC implemented an integrated microbial control and treatment optimization program.

Program Development

The initiative began with a comprehensive risk assessment of the crude oil network, benchmarking each segment against global best practices. The network was divided based on operational parameters, including:

- Temperature regimes
- Salinity and chemical environment
- Hydrogen sulfide (H₂S) presence
- Sulfate levels
- Fluid velocity
- Microbial cell counts (cells/mL)

This categorization enabled targeted, data-driven decision-making for MIC mitigation.

Risk Categorization and Benchmarks

Based on the assessment, ADNOC divided the network as follows:

Risk Categorization	Network Segment
No Risk	Wellhead and transfer lines to stations Temperature: 60-100°C High salinity: >250,000 mg/L H ₂ S: >0.2 bar Sulfates: 20-300 mg/L Velocity: >3 m/s Bacterial count: <100 cells/mL
High Risk	Crude oil export network Temperature: <60°C Low salinity: ~60,000 mg/L No H ₂ S Sulfates: 20-300 mg/L Velocity: 1-2 m/s Bacterial count: >1,000 cells/mL

MIC Treatment and Optimization

In low-risk segments, MIC treatment was deemed unnecessary and stopped, immediately saving over 2,400 drums of biocide per year (equivalent to USD \$2 million annually). For high-risk segments, the biocide treatment regimen was revamped and optimized:

- Dosage reduced from 500 ppm to 300 ppm
- Batch treatment duration cut from 5 hours to 3 hours
- Annual savings of 1,490 drums of biocide (USD \$1.05 million)

Overall, the program resulted in a remarkable 81% OPEX saving, exceeding \$3 million annually, while maintaining the integrity of the crude oil network.

Implementation Blueprint

ADNOC's success hinged on several best practices:

- Continuous data collection and microbial monitoring
- Regular risk reassessment and network segmentation
- Custom biocide regime development, rotating chemical agents to prevent resistance buildup
- Ongoing benchmarking with global leaders

Results and Impact

Not only did ADNOC's initiative drastically lower costs, but it also:

- Protected network integrity and prevented unplanned failures
- Reduced environmental footprint through minimized chemical use and emissions
- Provided a scalable model for other operators to replicate
- Enhanced operational sustainability and stakeholder confidence

DISCUSSION: INTEGRATING MIC OPTIMIZATION WITH BROADER CORROSION MANAGEMENT

The ADNOC case highlights the power of targeted, data-driven MIC control within a comprehensive corrosion management program. Several lessons emerge:

- Holistic Assessment: Integrating chemical, operational, and biological data yields the best outcomes.
- Adaptive Strategies: Regular reassessment and flexibility are vital, given the evolving nature of oilfield microbial communities.
- Cost-Benefit Focus: Optimization not only reduces OPEX but also improves asset longevity and sustainability.
- Knowledge Sharing: Disseminating best practices, such as those pioneered by ADNOC, can elevate industry standards worldwide.

CONCLUSIONS

Corrosion, in its many forms, continues to challenge oil and gas producers. However, meticulous program design, robust microbial monitoring, and dynamic risk management—exemplified by ADNOC's MIC

optimization initiative—demonstrate that significant reductions in both operational cost and environmental impact are achievable. As the industry evolves toward lower-carbon and more sustainable futures, such scientific and operational excellence will underpin asset integrity and responsible resource stewardship.

RECOMMENDATIONS FOR FUTURE OPERATIONS

- Maintain ongoing research into novel biocides and real-time microbial monitoring technologies.
- Encourage cross-industry collaboration for benchmarking and data sharing.
- Expand training for field personnel on the complexity of MIC and adaptive control strategies.
- Integrate corrosion management with broader sustainability and decarbonization goals.

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GLOSSARY OF TERMS

MIC	–	Microbial Induced Corrosion
OPEX	–	Operating Expenditure
SRB	–	Sulfate-Reducing Bacteria
APB	–	Acid-Producing Bacteria
Batch Treatment	–	A method of applying chemicals (such as biocides) in discrete doses over a specified period
Biocide	–	Chemical agent used to destroy harmful microorganisms
Corrosion Inhibitor	–	Chemical used to slow or prevent corrosion on metal surfaces
Cathodic Protection	–	A technique to prevent corrosion by making the protected metal the cathode of an electrochemical cell
Pigging	–	The process of cleaning pipelines using devices called ‘pigs’
Uniform Corrosion	–	Corrosion occurring evenly across a metal surface
Pitting Corrosion	–	Localized corrosion resulting in small, deep holes in metal
Crevice Corrosion	–	Corrosion that occurs in shielded areas such as joints or under deposits
Galvanic Corrosion	–	Corrosion resulting from electrical contact between dissimilar metals in the presence of an electrolyte
Erosion Corrosion	–	Corrosion accelerated by high-velocity fluid flow and particulates
Stress Corrosion Cracking (SCC)	–	Cracking caused by combined tensile stress and corrosive environment
Biofilm	–	A matrix produced by microorganisms that enables them to adhere to surfaces and accelerate corrosion
Risk Assessment	–	Evaluation of potential threats to asset integrity
Benchmarking	–	Comparing processes or performance metrics with industry best practices

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